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Playbook / Guide for ITAP Hackathon (University Series)

Purpose of the Hackathon

The Hackathon (University Edition) is designed to foster innovation, collaboration, and skill-building among university students while promoting Lockheed Martin's mission and values. It will provide:

1. **An opportunity for students to develop solutions** for real-world challenges within a competitive and supportive environment.
2. **Hands-on experience** in technology and teamwork, preparing participants for future professional opportunities.
3. **Visibility and outreach** for Lockheed Martin, showcasing our commitment to mentorship, education, and innovation.

Key Features of the Hackathon

1. **Challenge Tracks**:
    - Include multiple tracks aligned with Lockheed Martin's areas of expertise (e.g., cybersecurity, AI, aerospace).
    - Offer tailored challenges to allow participants to choose based on interest or skill.
2. **Team Formation and Mentorship**:
    - Provide a platform for participants to form teams or join pre-organized groups.
    - Assign Lockheed Martin mentors to guide teams throughout the hackathon.
3. **Location (venue) and Promotion**:
    - Host planning meetings with university representatives to secure venues.
    - Use media and communication tools to promote the event to universities and students, ensuring broad outreach and engagement.
4. **Judging and Awards**:
    - Establish a panel of judges comprising Lockheed Martin experts and university faculty.
    - Offer awards for categories like best innovation, most practical solution, and best teamwork.
5. **Networking Opportunities**:
    - Create spaces for students to interact with Lockheed Martin professionals, university leaders, and fellow participants.

Steps to Organize the Hackathon

1. **Define Objectives and Themes**:
    - Establish clear goals and select themes based on Lockheed Martin's priorities.
    - Collaborate with university faculty to align challenges with students' curriculum.
2. **Set Up Registration Platform**:
    - Create an online portal or use a service like WordPress to manage participant sign-ups.
    - Include fields for team information, preferred challenge tracks, and skill levels.
3. **Host Meetings with Universities**:
    - Secure location (venue) for the hackathon and discuss logistical needs with university representatives.
    - Build strong partnerships with university to ensure their support and participation.
4. **Promote the Event**:
    - Leverage media channels and communication tools to spread awareness among students and faculty.
    - Use social platforms, email campaigns, and university partnerships to maximize outreach.
5. **Deliver Challenges**:
    - Clearly present the challenges and set up communication channels for team collaboration.
    - Provide necessary tools and resources for participants to complete tasks efficiently.
6. **Run the Hackathon**:
    - Facilitate team progress with check-ins and support from mentors.
    - Ensure smooth operations, resolve technical issues, and keep participants engaged.
7. **Judging and Awards Ceremony**:
    - Conduct a final judging session to evaluate solutions based on predefined criteria.
    - Announce winners and distribute awards to celebrate achievements.
8. **Feedback and Follow-Up**:
    - Collect feedback from participants and mentors to improve future events.
    - Provide opportunities for continued engagement, such as internships or collaboration with Lockheed Martin teams.

Hackathon Playbook: AI Flashcards Project & Logistics

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Objective
Default Project:

Participants will be tasked with creating a flashcard application that not only leverages a provided API key to generate flashcards based on user-inputted prompts, but also features a visually appealing and responsive frontend. The goal of the app is to allow users to write a prompt and create flashcards on that subject using the response from the API endpoint, and a well-designed interface will be crucial in making this process intuitive and user-friendly. By combining functional backend capabilities with a polished and interactive frontend, teams will need to demonstrate their ability to craft a seamless and engaging user experience.

- **Goa**l: Create a flashcard application that generates flashcards based on user-inputted prompts using a provided API key.
- **Core Tasks:**
    - **Backend:** Leverage the API key to generate flashcards based on user prompts.
    - **Frontend:** Design a visually appealing and responsive interface that allows users to input prompts and view generated flashcards.
    - **User Experience:** Combine functional backend capabilities with a polished and interactive frontend to create a seamless and engaging user experience. -
    - **Key Features:** Allow users to write a prompt, generate flashcards on that subject using the API endpoint response, and interact with the application in an intuitive and user-friendly way.

Open Project:

In addition to the flashcard application challenge, participants are also encouraged to think outside the box and propose their own innovative project ideas that align with the AI theme of the hackathon. This is an opportunity for teams to showcase their creativity and expertise by developing a unique solution that showcases the potential of AI, and we welcome any idea that pushes the boundaries of what is possible with artificial intelligence.

Hackathon Timeline (10 Hours) (Subject to change)

Pre-Event Setup

- **7:00 AM - 7:30 AM**:
    - Teams arrive onsite, check in, and set up workspaces.
    - ITAP team ensures all technical setups (laptops, internet, website access) are functioning correctly.
    - Mentors and volunteers are briefed on their roles.

- **7:30 AM - 8:00 AM**:
  - **Orientation & Kickoff**:
    - Welcome participants and introduce the hackathon objectives.
    - Outline the judging criteria and website integration process.
    - Provide a short walkthrough of team responsibilities and the schedule.

Hackathon Development Schedule

- **8:00 AM - 11:30 AM**:
  - **Phase 1: Ideation and Development**:
    - Teams finalize their project ideas and begin working
    - Mentors provide guidance on connecting the API and reading data from a response.

- **11:30 AM - 12:30 PM**:
  - **Lunch Break**:
    - Provide a hot or buffet-style lunch to give participants time to refuel.

- **12:30 PM - 2:45 PM**:
  - **Phase 2: teams will continue to work on their ideas**:
    - Mentors assist with debugging and troubleshooting.

- **2:45 PM - 3:00 PM**:
  - **Wrap-Up and Submissions**:
    - Teams submit their final projects on the website for judging.

Judging and Awards

- **3:00 PM Onwards**:
  - Judges evaluate submissions and deliberate on rankings independently or in a separate review session.
  - Results are shared via follow-up communication or live announcements (depending on available time after 3 PM).

Judging Criteria

Scores are assigned based on specific categories, with weighted scoring applied for fairness:

Category	Weight	What to Evaluate
Technical Execution	30%	How well the team processed the API return data (If the response is relevant to the prompt)
Creativity and Innovation	25%	The originality of the idea and approach. Appeling to the eye
Impact and Relevance	20%	How meaningful or impactful the project is to its target audience.
Presentation and Storytelling	15%	Clarity, professionalism, and engagement during the demo.
User Experience	10%	How intuitive and responsive the visualizations and chatbot are.

Example Judging Scores

Here's an example of how scores could look for three teams after testing their contributions:

Team	Execution (30%)	Creativity (25%)	Impact (20%)	Presentation (15%)	UX (10%)	Final Score
Team A	**8 (2.4)**	**9 (2.25)**	**7 (1.4)**	**6 (0.9)**	**8 (0.8)**	**7.75**
Team B	**9 (2.7)**	**8 (2.0)**	**9 (1.8)**	**7 (1.05)**	**6 (0.6)**	**8.15**
Team C	**7 (2.1)**	**10 (2.5)**	**8 (1.6)**	**8 (1.2)**	**7 (0.7)**	**8.1**

Results:

- **1st Place:** Team B (8.15 points).
- **2nd Place:** Team C (8.1 points).
- **3rd Place:** Team A (7.75 points).

Final Presentations:

- Each team gives a 5–10-minute demo, showing their Flashcard App, visualizations, and chatbot functionality live on the website.
- Judges interact with each team's chatbot and test dynamic updates.

Scoring:

- Judges score each team based on the weighted criteria.
- Scores are tallied, and the top three teams are ranked.

Note: Teams can create their own names

Roles Needed for ITAP to Execute - 10-person ITAP team with volunteers:

1. **Hackathon Coordinators (2 ITAP Members – Lead, Co-Lead and support team) • Responsibilities**:
    - Oversee all logistics, team assignments, and overall event flow.
    - Coordinate resources like laptops, internet, and scheduling.
    - Serve as the point of contact for escalations or urgent issues.

2. **Technical Mentors (4 ITAP Members + Volunteers) • Responsibilities**:
    - Support participants in Connecting the API, cleaning data, and integrating visualizations into the website.
    - Host brief tutorials on tools (e.g., Python, JavaScript APIs) to onboard students quickly.
    - Troubleshoot common challenges (e.g., handling JavaScript-loaded content, rate limits).
    - **Volunteers' Role**: Pair with ITAP mentors to assist participants by answering basic questions or offering guidance on simpler tasks.

3. **Judging Panel (3 ITAP Members + 2 Volunteers) • Responsibilities**:
    - Evaluate projects based on the predefined criteria (execution, creativity, impact, presentation, and user experience).
    - Test each team's contributions live on the central website.
    - Assign scores and deliberate to rank 1st, 2nd, and 3rd places.
    - **Volunteers' Role**: Act as guest judges or industry experts, bringing fresh perspectives to evaluations.

4. **Website Management Team (2 ITAP Members)**
    - **Responsibilities**: **(Before the Hackathon)**
        - Build the initial website framework with designated sections.
            - ITAP prepares and hosts the website, with designated sections for each team's contributions.
            - Judges receive the judging criteria and familiarize themselves with the evaluation platform.

- Set up backend endpoints to allow teams to upload and dynamically update their data.
- Monitor the website during the event to ensure dynamic updates reflect correctly.
- **During the Hackathon**:
  - Teams upload their progress to the website as they complete Api connection (etc..).
  - Judges monitor live submissions and test functionality in real time.
- **Testing Criteria**:
  - **Application Testing**: Judges evaluate the API response(e.g., accuracy, relativeness to the prompt) by running tests or viewing uploaded results.
  - **Visualization & Chat Box**: Judges test chatbot responses and visualize data in the flashcards

5. Presentation Coaches (2 ITAP Members)

- **Responsibilities**:
  - Help teams refine their project pitches and demo presentations.
  - Provide tips on storytelling, emphasizing the project's impact and innovation.
  - Offer advice on creating engaging and professional visuals for the live demo.

6. Communication Lead (1 ITAP Member + Volunteers)

- **Responsibilities**:
  - **Pre-Hackathon**:
    - Create and design promotional flyers to be shared by the university to attract student participants.
    - Draft email communications and announcements to ensure smooth registration and clarity on event details.
    - Event details, timeline, and judging criteria.
    - Resources like tool tutorials, example projects, and links to documentation.
    - Create separate channels for general announcements, mentor support, and team discussions. Set up a dedicated communication platform (e.g., Slack, Microsoft Teams, or Discord).

  - **Day of the Hackathon**:
    - Manage real-time announcements, including schedule updates and reminders.
    - Coordinate messaging to keep participants informed throughout the event.
  - **Post-Hackathon**:

        - Develop and distribute post-event communications, including sharing results, project highlights, and thank-you messages.
        - Create and distribute a participant feedback survey to gather insights for future events.
    - **Volunteer Support Role**:
        - Assist in distributing flyers and managing announcements.

Meeting Schedule for Hackathon Planning

1. **Weekly Meetings** *(Now - Mid-May)*:
    - Brainstorm ideas, finalize logistics, and prepare for the university presentation.
    - Assign tasks for presentation materials and team roles.
2. **Biweekly Meetings** *(Late May - Mid-August)*:
    - Monitor progress on recruitment, promotional activities, and any training sessions.
    - Ensure logistical planning stays on track (e.g., platform setup, judging criteria, etc.).
3. **Weekly Meetings** *(Late August - September 27)*:
    - Focus on finalizing all hackathon details, such as participant guidelines, team assignments, and ensuring resources (judges, mentors, platforms, etc.) are confirmed.
4. **Hackathon Day** *(September 27)*
    - Host the event, ensuring smooth execution of all phases: orientation, development, judging, and awards.
    - Gather participant feedback at the end of the day for future improvements.